# Confidence-Weighted Multi-Scale Rolling Forecasting of Urban Taxi Demand with GraphSAGE-Based Spatial Refinement

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# Abstract

Accurate urban taxi demand forecasting is essential for intelligent transportation management, fleet dispatching, and congestion mitigation. However, this task remains challenging because demand patterns exhibit strong temporal non-stationarity, multi-scale periodicity, and complex spatial dependency across city regions. To address these issues, this study proposes a two-stage framework for hourly taxi demand prediction over New York City Taxi Zones, integrating multi-scale temporal modeling, uncertainty estimation, and graph-based spatial refinement. In the first stage, a persistent per-zone forecasting model is developed to capture temporal dynamics at four different scales: hourly, daily, weekly, and monthly. The architecture combines GRU, LSTM, and Transformer components to learn short-term fluctuations and long-range temporal regularities within a unified multi-branch design. Monte Carlo Dropout is further employed to estimate predictive uncertainty for each zone. In the second stage, the base predictions are treated as priors and refined through a GraphSAGE-based residual learning module constructed on an origin-destination flow graph. To improve robustness, a confidence-weighting mechanism is introduced by combining historical sufficiency, model stability, and neighborhood consistency, allowing unreliable zones to contribute less during graph refinement. The overall system is evaluated under a rolling forecasting protocol that simulates real-world hourly prediction updates. Experimental results indicate that the proposed graph refinement stage reduces prediction error relative to the temporal baseline and is especially beneficial for zones with sparse history or high predictive variance. In addition, the proposed method improves spatial coherence among neighboring zones and provides a more stable forecasting process under dynamic urban conditions. Overall, the framework demonstrates that combining multi-scale sequence learning with confidence-aware graph neural refinement offers an effective, robust, and extensible solution for spatiotemporal urban taxi demand forecasting problems.

# Acknowledgements

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# Chapter 1

# Introduction

## 1.1 Background and Motivation

Urban mobility forecasting has become a central problem in intelligent transportation systems because accurate short-term demand estimates influence vehicle dispatching, idle cruising reduction, passenger waiting time, congestion management, and the overall operational efficiency of city-scale transport services.[1] In large metropolitan areas, taxi demand is not only highly dynamic but also strongly heterogeneous across space and time.[2] Different urban regions exhibit distinct activity rhythms shaped by commuting peaks, nightlife, commercial density, tourism, and residential land use. At the same time, the temporal behaviour of demand is rarely governed by a single periodic pattern. Instead, it reflects a mixture of short-term fluctuation, daily repetition, weekly regularity, and longer-range seasonal tendencies. These properties make hourly taxi demand forecasting a difficult spatiotemporal learning task rather than a conventional univariate time-series regression problem.[3]

The challenge becomes more pronounced when the forecasting objective is defined at the level of individual Taxi Zones. Fine-grained zone-level prediction is practically useful because dispatch systems, relocation strategies, and local service balancing all depend on geographically resolved demand estimation.[4] However, a disaggregated formulation also introduces additional statistical difficulty. Some zones generate abundant historical observations, while others are comparatively sparse or irregular. In low-activity regions, the observed series may contain many zero or near-zero counts interspersed with occasional surges, which makes the learning problem noisy and unstable due to sparsity and uncertainty in travel demand data [5]. In high-activity districts, by contrast, the model must handle larger amplitude fluctuations and sharper temporal transitions, requiring strong capability in capturing complex spatial-temporal dependencies [6]. A forecasting framework that performs well across all zones therefore needs to balance sensitivity to local temporal dynamics with robustness to data sparsity and uncertainty.

A second source of complexity lies in the spatially coupled nature of urban travel demand. Taxi zones do not evolve independently. Neighbouring regions, or regions connected by strong origin-destination flows, often exhibit correlated demand patterns because travel activity propagates through the city via commuting corridors, business districts, airports, entertainment clusters, and residential catchment areas.[1] A purely per-zone temporal model can learn local history effectively, but it may fail to correct errors that arise when neighbouring zones move coherently or when a zone's local history is insufficient to explain its next-hour demand. This is particularly problematic under rolling forecasting conditions, where predictions must be refreshed hour by hour and small local errors can accumulate into spatially inconsistent outputs. Capturing such dependence requires a representation that can use structural information beyond the isolated time series of a single region.

Beyond technical performance, this problem also has wider urban significance. More reliable demand forecasting can support better vehicle allocation and improve the spatial matching between supply and expected local demand [7], [8]. It can also help reduce empty cruising or deadhead mileage, thereby improving operational and potentially energy efficiency [9]. In a data-scarce or highly variable urban environment, a forecasting system must therefore be judged not only by average error metrics but also by its stability, interpretability, and ability to remain useful across heterogeneous operational contexts. This emphasis on operational realism gives the study both academic relevance and applied value.

## 1.2 Problem Definition and Research Challenges

Recent research on traffic and mobility prediction has increasingly moved toward spatiotemporal architectures that combine temporal sequence modelling with graph-based spatial learning.[10] Temporal models such as recurrent neural networks and transformer variants are effective for extracting sequential dependencies**info15090517rnn_evolution_2024**, whereas graph neural networks offer a principled way to propagate information across related regions[11]. Yet the The integration of these components is not straightforward. End-to-end spatiotemporal models can be highly effective for traffic forecasting because they jointly model spatial and temporal dependencies, and representative architectures such as DCRNN and Graph WaveNet have achieved strong results in this setting [12], [13]. However, such graph-based deep models are often difficult to interpret, which limits their transparency in operational use [14]. In practical city-scale forecasting, there is therefore value in separating the problem into an initial temporal prediction stage and a subsequent spatial correction stage. This type of decoupled design is also supported by recent work showing that different latent traffic signals can be handled more effectively when spatial diffusion and inherent temporal components are modeled separately [15]. Such a design allows the first component to focus on modelling zone-specific temporal regularity while the second component exploits graph structure to refine predictions in a more modular and controlled manner.

Another important issue is uncertainty. Most prediction systems output a single point estimate, but for urban demand forecasting this is frequently insufficient, since recent studies have shown that spatiotemporal travel demand prediction should explicitly quantify predictive uncertainty rather than rely only on deterministic outputs [5], [16]. Prediction reliability varies significantly across time and across regions. A confident prediction in a historically dense, stable zone should not be treated in the same way as a volatile estimate produced for a sparse zone with weak historical support [5]. If uncertainty can be quantified explicitly, it can be used to guide later stages of the model and to reduce the influence of unreliable nodes during learning. Therefore, uncertainty is not merely an auxiliary diagnostic quantity; it can also serve as a useful signal for weighting, calibration, and robustness enhancement.

From a research perspective, the problem addressed in this thesis sits at the intersection of spatiotemporal forecasting, graph neural networks for transport modelling, uncertainty-aware prediction, and deployment-oriented rolling inference [13]. Its novelty does not rely on any single component in isolation. GRU, Monte Carlo Dropout, and GraphSAGE are all established methods in sequence modelling, uncertainty estimation, and inductive graph representation learning, respectively [17], [18], [19]. The contribution instead lies in how these components are combined into a coherent forecasting pipeline for zone-level hourly taxi demand.

At the same time, the current project setting presents several important methodological challenges. First, the graph refinement stage is currently trained and evaluated on the same hourly batch, which may inflate the apparent gain of graph-based correction and weaken claims about cross-temporal generalisation. Second, although edge weights are constructed and preserved in the data pipeline, they are not yet explicitly exploited inside the `SAGEConv` message-passing operator. Third, the temporal forecasting framework now supports both persistent checkpoint reuse and lightweight incremental updating; however, in the default rolling configuration, the persistent version is still used unless the script is manually switched to the incremental manager. Consequently, the system already contains a practical path toward online adaptation, but this capability is not yet the default experimental setting. These issues should not be viewed simply as implementation defects. Instead, they delineate the real methodological boundary of the current study and provide a clear basis for the experimental analysis, ablation design, and future improvements discussed in this thesis.

## 1.3 Research Rationale and Proposed Framework

The present project is motivated by precisely these difficulties. It addresses hourly taxi demand prediction for New York City Taxi Zones through a two-stage, confidence-weighted, multi-scale forecasting framework. According to the project blueprint, the system is organised around three linked ideas: first, each zone is predicted with a persistent multi-branch temporal model that captures information at hourly, daily, weekly, and monthly scales; second, predictive uncertainty is estimated through Monte Carlo Dropout and fused with other reliability indicators to form a confidence score [18]; third, the initial predictions are refined on a graph built from origin-destination flow relationships using a GraphSAGE residual learning module [19]. The overall pipeline is executed in a rolling fashion, producing hour-by-hour forecasts, metrics, and output files that simulate a realistic operational deployment setting.

The choice of a multi-scale temporal backbone is central to the rationale of this work. Urban taxi demand is inherently hierarchical in time. Short-term behaviour captures immediate fluctuations and local momentum. Daily periodicity reflects repeated within-day routines such as morning commute and evening return flows. Weekly periodicity encodes the distinction between weekdays and weekends, and longer temporal windows can provide a broader view of persistent demand level and slow trend evolution. A model built on a single temporal granularity often struggles to preserve all of these signals simultaneously. In this project, the temporal stage therefore uses four branches corresponding to 1-hour, 1-day, 1-week, and 1-month contexts. The implementation combines LSTM encoders for daily and weekly windows, a Transformer-based encoder for the longer monthly window, and a GRU-driven output pathway for the short-term branch, with the fused trend representation used to support final next-step prediction. This design reflects an explicit modelling assumption: different temporal horizons contribute complementary predictive information, and their joint representation is more suitable for robust zone-level forecasting than any single-scale view alone [20], [21].

Equally important is the decision to adopt a persistent per-zone training strategy rather than a monolithic city-wide temporal model. In the current implementation, each zone maintains its own checkpointed forecasting model, allowing training to occur once and then be reused during subsequent rolling steps. This persistent formulation has two practical advantages. First, it respects local heterogeneity by letting each zone learn its own temporal dynamics without forcing all regions into a single parameter space. Second, it reduces unnecessary retraining cost during rolling evaluation, which is useful when the system is intended to mimic hourly forecasting updates. At the same time, the project explicitly recognises that persistent checkpoint reuse is not the only available operating mode. The repository also provides an incremental training variant in which newly arrived time windows trigger lightweight model updates. However, in the present rolling script configuration, the persistent manager remains the default setting, so online adaptation is supported by the codebase but not enabled by default.

The uncertainty component further differentiates this framework from a standard multi-scale recurrent model. Instead of treating prediction variance as a post hoc statistic, the system uses Monte Carlo Dropout to generate repeated stochastic forward passes and estimate predictive standard deviation [18]. These uncertainty estimates are then integrated with two additional reliability signals: a prior score based on the historical richness of each zone, and a neighbourhood consistency score derived from agreement with graph neighbours. The resulting confidence value is not only recorded as an interpretive measure, but also used downstream to weight graph-based learning. In real demand data, zones with limited history, unstable predictions, or poor agreement with surrounding structure are precisely the cases in which the model should be more cautious. Confidence weighting operationalises that caution mathematically and links uncertainty estimation to model robustness.

The graph refinement stage addresses the main limitation of independent temporal forecasting: lack of spatial correction. After the base model generates per-zone next-hour predictions, the second stage treats these values as priors and learns residual corrections over an origin-destination flow graph. The use of GraphSAGE is notable here because the model is not asked to predict demand from scratch on the graph. Instead, it learns how much the initial temporal estimate should be adjusted once spatial context is taken into account. This residual perspective lowers the burden placed on the graph model and makes the refinement process easier to interpret because improvements can be understood as structured corrections rather than opaque end-to-end remapping [19].

The use of an origin-destination flow graph rather than a purely geographic adjacency graph is also theoretically meaningful. A mobility system is shaped not only by physical proximity but by functional interaction. Two zones may be geographically near yet weakly related in taxi demand, while distant zones can exhibit strong dependence if substantial travel flow connects them. By constructing adjacency from an OD flow matrix, the project attempts to encode actual travel coupling rather than crude map neighbourhood alone. This choice aligns with the practical intuition that mobility demand propagates along usage structure, not merely spatial contiguity [1], [22].

The rolling evaluation protocol is another defining feature of this study. Instead of training once and reporting static test results, the pipeline advances target time step by time step, recalculating predictions and metrics at each hour. This setup better resembles how forecasting systems are used in deployment, where the model must repeatedly produce near-future estimates as new time points arrive. Rolling evaluation is important not only for realism but also for diagnosis. It reveals whether performance is stable over time, whether certain hours are systematically harder than others, and whether the spatial refinement stage consistently improves over the temporal baseline.

## 1.4 Research Aim and Questions

Against this background, the aim of this thesis is to investigate whether a two-stage architecture can improve hourly urban taxi demand forecasting by combining local multi-scale temporal modelling with confidence-aware graph refinement.

More specifically, this thesis addresses the following research questions:

1. Can a per-zone multi-scale recurrent architecture provide robust next-hour predictions across heterogeneous Taxi Zones?
2. Can uncertainty-derived confidence scores improve the stability of spatial refinement by weighting graph learning according to node reliability?
3. Does a GraphSAGE residual correction stage produce more accurate and spatially coherent forecasts than the temporal baseline alone under a rolling evaluation setting?

## 1.5 Contributions of This Thesis

Accordingly, the main contributions of this work can be summarised as follows:

1. It develops a multi-scale temporal forecasting framework for hourly Taxi Zone demand that combines GRU, LSTM, and Transformer components across four temporal horizons.
2. It incorporates Monte Carlo Dropout-based uncertainty estimation into a broader confidence fusion mechanism that also accounts for historical sufficiency and neighbourhood consistency.
3. It introduces a GraphSAGE-based residual refinement stage over an origin-destination flow graph to correct temporal predictions using spatial structure.

4. It implements the whole method within a rolling prediction pipeline with persistent checkpoint reuse as the default setting, while also supporting an incremental update variant for improved rolling-stage stability.

Together, these components form a practical and extensible foundation for urban demand forecasting in settings where temporal complexity, spatial coupling, and uncertainty must all be considered simultaneously.

## 1.6 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 reviews related work on urban demand forecasting, multi-scale temporal sequence modelling, graph neural networks, and uncertainty-aware prediction. Chapter 3 formalises the problem setting and presents the proposed two-stage framework in detail, including temporal architecture, confidence construction, graph refinement, and rolling inference logic. Chapter 4 describes the implementation pipeline, data preparation process, key classes, and engineering decisions reflected in the codebase. Chapter 5 evaluates the temporal baseline and graph-refined model under rolling forecasting conditions and discusses performance patterns. Finally, Chapter 6 concludes the thesis with a discussion of strengths, limitations, and future directions, including better generalisation protocols, explicit edge-weight usage, and more systematic evaluation of the incremental online adaptation mechanism already provided by the repository.

# Chapter 2

# Literature Review

## 2.1 Introduction to the Literature Context

Urban mobility forecasting has become a central problem in intelligent transportation systems because accurate short-term demand estimates influence vehicle dispatching, idle cruising reduction, passenger waiting time, congestion management, and the overall operational efficiency of city-scale transport services [23], [24]. In large metropolitan areas, taxi demand is not only highly dynamic but also strongly heterogeneous across space and time [2], [25]. Different urban regions exhibit distinct activity rhythms shaped by commuting peaks, nightlife, commercial density, tourism, and residential land use [25]. At the same time, the temporal behaviour of demand is rarely governed by a single periodic pattern. Instead, it reflects a mixture of short-term fluctuation, daily repetition, weekly regularity, and longer-range seasonal tendencies, which makes hourly taxi demand forecasting a difficult spatiotemporal learning task rather than a conventional univariate time-series regression problem [13], [20].

The challenge becomes more pronounced when the forecasting objective is defined at the level of individual Taxi Zones. Fine-grained zone-level prediction is practically useful because dispatch systems, relocation strategies, and local service balancing all depend on geographically resolved demand estimation [24], [26]. However, a disaggregated formulation also introduces additional statistical difficulty. Some zones generate abundant historical observations, while others are comparatively sparse or irregular. In low-activity regions, the observed series may contain many zero or near-zero counts interspersed with occasional surges, making the learning problem noisy and unstable [5]. In high-activity districts, by contrast, the model must handle stronger amplitude variation and potentially sharper peak transitions, requiring effective modelling of complex spatial-temporal dependence [6], [27]. A forecasting framework that performs well across all zones therefore needs to balance sensitivity to local temporal dynamics with robustness to data sparsity and uncertainty.

A second source of complexity lies in the spatially coupled nature of urban travel demand. Taxi zones do not evolve independently. Neighbouring regions, or regions connected by strong origin-destination flows, often exhibit correlated demand patterns because travel activity propagates through the city via commuting corridors, business districts, airports, entertainment clusters, and residential catchment areas [22], [28]. A purely per-zone temporal model can learn local history effectively, but it may fail to correct errors that arise when neighbouring zones move coherently or when a zone's local history is insufficient to explain its next-hour demand. This is particularly problematic under rolling forecasting conditions, where predictions must be refreshed hour by hour and small local errors can accumulate into spatially inconsistent outputs. Capturing such dependence therefore requires a representation that can use structural information beyond the isolated time series of a single region [13].

For this reason, recent research on traffic and mobility prediction has increasingly moved toward spatiotemporal architectures that combine temporal sequence modelling with graph-based spatial learning [13], [29]. Temporal models such as recurrent neural networks and transformer variants are effective for extracting sequential dependencies, whereas graph neural networks offer a principled way to propagate information across related regions [13]. Yet the integration of these components is not straightforward. End-to-end spatiotemporal models can be powerful, but they are often computationally heavy, difficult to interpret, and less flexible when the temporal and spatial signals differ in quality across nodes [14], [30]. In practical city-scale forecasting, there is value in separating the problem into an initial temporal prediction stage and a subsequent spatial correction stage. Such a design allows the first component to focus on modelling zone-specific temporal regularity while the second component exploits graph structure to refine predictions in a controlled and interpretable manner [15].

Another important issue is uncertainty. Most prediction systems output a single point estimate, but for urban demand forecasting this is frequently insufficient, since recent work has shown that spatiotemporal travel demand prediction benefits from explicitly quantifying predictive uncertainty rather than relying only on deterministic outputs [5], [16]. Prediction reliability varies significantly across time and across regions. A confident prediction in a historically dense, stable zone should not be treated in the same way as a volatile estimate produced for a sparse zone with weak historical support [5]. If uncertainty can be quantified explicitly, it can guide later stages of the model and reduce the influence of unreliable nodes during learning. Therefore, uncertainty is not merely an auxiliary diagnostic quantity; it can also serve as a useful signal for weighting, calibration, and robustness enhancement.

The present project is motivated by precisely these difficulties. It addresses hourly taxi demand prediction for New York City Taxi Zones through a two-stage, confidence-weighted, multi-scale forecasting framework. According to the project blueprint, the system is organised around three linked ideas: first, each zone is predicted with a persistent multi-branch temporal model that captures information at hourly, daily, weekly, and monthly scales; second, predictive uncertainty is estimated through Monte Carlo Dropout and fused with other reliability indicators to form a confidence score [18]; third, the initial predictions are refined on a graph built from origin-destination flow relationships using a GraphSAGE residual learning module [19]. The overall pipeline is executed in a rolling fashion, producing hour-by-hour forecasts, metrics, and output files that simulate a realistic operational deployment setting.

This chapter therefore reviews the literature in six connected parts. First, it examines traditional approaches to urban demand forecasting and their limitations. Second, it reviews deep temporal sequence models and explains why multi-scale learning became important. Third, it surveys spatial modelling approaches, especially graph neural networks, and discusses the relevance of GraphSAGE-style aggregation. Fourth, it considers hybrid spatiotemporal architectures and the distinction between end-to-end fusion and two-stage refinement. Fifth, it reviews uncertainty estimation and confidence-aware learning. Finally, it discusses rolling forecasting, persistent training, and incremental adaptation, which are central to practical deployment. Throughout the chapter, the review will relate prior work back to the needs of the current project and identify the methodological gaps that the thesis should address.

## 2.2 Traditional Urban Demand Forecasting Methods

The earliest literature on transportation demand forecasting was dominated by statistical and econometric models. Classical methods included autoregressive integrated moving average models, seasonal ARIMA variants, exponential smoothing, Poisson and negative binomial regression, and state-space formulations [31], [32]. These methods remain attractive because they are interpretable, computationally light, and theoretically well understood. In settings with stable seasonality and relatively smooth temporal structure, such models can provide strong baselines. For aggregate traffic or passenger series, they often perform reasonably well when the forecasting horizon is short and the signal is dominated by linear dependence [31].

However, taxi demand forecasting at the zone level quickly exposes the limits of such models. First, linear autoregressive methods assume relatively simple dependence structures and struggle when multiple temporal scales interact nonlinearly. Second, they are poorly suited to sparse or highly intermittent zones where many observations are zero or near-zero [5]. Third, classical models generally treat each spatial unit independently unless augmented with hand-designed exogenous terms or multivariate extensions, which become difficult to scale in city-wide settings [13], [33]. Fourth, the assumptions required for stationarity, residual independence, or parametric count structure are often violated by urban mobility data [32]. These weaknesses motivated the field to adopt machine learning methods capable of handling high-dimensional nonlinearity more flexibly.

Before deep learning became dominant, methods such as support vector regression, random forests, gradient boosting, and k-nearest-neighbour regression were frequently applied to transport prediction [34], [35]. These models could capture nonlinear feature interactions better than traditional statistical methods, especially when provided with engineered inputs such as hour-of-day, day-of-week, weather indicators, holiday flags, and lagged demand values [32], [35]. Nevertheless, they still depended heavily on feature engineering and did not solve the core representation problem of spatiotemporal dependence. A random forest may capture nonlinear interactions among static covariates, but it does not naturally model long temporal memory or structured spatial propagation. Similarly, support vector methods can be effective on carefully prepared datasets but do not scale elegantly to multi-zone dynamic graphs with rolling re-estimation requirements [13], [34].

A further limitation of both statistical and early machine learning approaches is that they often produce point forecasts without any integrated uncertainty mechanism. In urban mobility applications, this is a significant gap. Demand variance is not homogeneous across zones or time periods, and decision systems can benefit greatly from knowing which predictions are stable and which are unreliable [5], [16]. Thus, even before discussing deep learning, the literature already points toward three essential requirements that simpler approaches do not jointly satisfy: nonlinear multi-scale temporal modelling, explicit spatial coupling, and reliability awareness [5], [13].

These limitations are directly relevant to the current project. The system is designed around zone-level demand prediction with mixed short- and long-term temporal patterns, OD-related spatial coupling, and uncertainty-aware graph refinement through confidence scores. This implies that purely statistical baselines, while still necessary for comparison, cannot be the final methodological destination of the thesis. Instead, they serve mainly as conceptual stepping stones illustrating why a deeper spatiotemporal architecture is justified.

## 2.3 Deep Temporal Sequence Models for Demand Forecasting

The deep learning literature on forecasting introduced a major shift by replacing manual feature engineering with learned representations. Recurrent neural networks became especially influential because they provide a natural mechanism for modelling sequential dependence. Standard RNNs can, in principle, use previous hidden states to summarise temporal context, but in practice they suffer from vanishing and exploding gradient problems when sequences are long [36]. This limitation led to the widespread adoption of Long Short-Term Memory networks and Gated Recurrent Units, both of which use gating mechanisms to regulate information flow and preserve relevant historical context over longer horizons [17], [37].

In transportation forecasting, LSTM-based models were soon applied to traffic speed, passenger flow, and taxi demand because they captured nonlinear temporal dependence more effectively than traditional autoregressive models [13], [38]. GRUs offered a lighter-weight alternative with fewer parameters and often similar empirical performance [39]. In urban demand settings, recurrent models are useful because they can absorb sequential fluctuations without requiring explicit manual design of every lag interaction. They also adapt relatively naturally to rolling one-step-ahead prediction tasks, which is important for hourly operational forecasting.

Yet the recurrent literature also exposed a key problem: a single recurrent branch is often insufficient to capture the coexistence of multiple temporal rhythms. Taxi demand does not evolve only according to the immediately preceding hour. There are within-day cycles, weekly regularities, event-driven deviations, and slower baseline shifts. A model trained only on a short recent history may miss recurrent weekly effects; a model trained on a very long history may dilute immediate local momentum. This tension drove the literature toward multi-scale or multi-resolution temporal modelling [20], [40].

Multi-scale temporal models attempt to process different context windows separately and then combine them into a unified predictive representation. Some methods use parallel recurrent branches fed with different lag lengths. Others separate local and global temporal streams, or introduce convolutional structures to summarise short-term patterns before recurrent integration [40], [41]. More recently, transformer-based encoders have been used to model long-range dependencies, since attention mechanisms can relate distant time points more directly than purely recurrent hidden-state propagation [41], [42]. In mobility prediction, multi-scale designs are attractive because they align closely with known human activity rhythms: hourly recency, daily routine, weekly schedule, and longer background trend [20], [25].

The current project follows this line of development quite explicitly. The temporal stage uses four scales: 1 hour, 1 day, 1 week, and 1 month. The implementation combines LSTM encoders for daily and weekly sequences, a Transformer-based pathway for the monthly context, and a GRU-centred output structure for short-term prediction. This makes the model neither a pure LSTM, pure GRU, nor pure Transformer architecture. Instead, it is a composite multi-branch design in which each component handles the timescale for which it is most suitable. Such a design is well supported by the literature because it reflects the widely recognised principle that no single temporal mechanism dominates all forecasting horizons equally [41], [42].

Another relevant theme in the sequence-model literature is the trade-off between local adaptation and global sharing. Many deep forecasting studies train a single city-wide model using all zones jointly, often with embeddings to represent region identity. This can improve data efficiency because parameters are shared across zones. However, shared models may also obscure strong local heterogeneity, especially when different regions exhibit substantially different temporal patterns [43], [44]. A highly active commercial district and a low-density peripheral zone may not benefit equally from identical temporal parameters. By contrast, per-zone models allow local dynamics to be learned independently but can suffer from limited data in sparse regions. The project adopts a per-zone temporal modelling approach that preserves local structure while later compensating for spatial dependence through graph refinement. This choice aligns with literature that emphasises the importance of respecting regional heterogeneity in mobility prediction, but it also invites comparison with shared or meta-learned temporal models as possible future improvements [45], [46].

A further issue concerns input scaling and zero inflation. Mobility count series often contain sharp spikes, long low-activity periods, and substantial imbalance across zones. In such settings, deep temporal models can become sensitive to amplitude disparities unless scaling is carefully handled. The framework uses MinMax scaling and dense hourly reconstruction with missing hours filled by zero. These are practical engineering decisions consistent with the broader forecasting literature, though they also introduce open questions. For example, MinMax scaling can be sensitive to extreme values, and zero-filling may blur the distinction between true absence and missing observation. These issues do not invalidate the approach, but they are the kinds of modelling assumptions that literature review should surface because they shape how results ought to be interpreted [5], [43].

Overall, the literature on deep sequence learning provides strong support for the project's temporal stage. It justifies the movement beyond classical time-series baselines, motivates the use of gated recurrent models, and explains the need for explicitly multi-scale architecture [41], [43]. At the same time, it suggests several comparative angles that the thesis could later exploit: single-scale versus multi-scale models, GRU-only versus hybrid designs, shared city-level versus per-zone models, and recurrent versus transformer-heavy long-context encoders.

## 2.4 Multi-Scale Modelling as a Distinct Research Theme

Although multi-scale sequence learning can be discussed under deep temporal modelling, it has become important enough in transportation prediction to warrant a separate treatment. The broader forecasting literature repeatedly shows that urban mobility contains superposed rhythms rather than one dominant cycle. Commuting activity imposes strong diurnal repetition, weekends introduce behavioural discontinuity, and slower structural shifts emerge from tourism, economic conditions, school terms, and seasonal patterns. Models that use only a fixed-length lag often end up overfitting one scale and under-representing another [20], [25].

One family of multi-scale approaches uses handcrafted lag groups. For instance, a model may include the last few hours, the same hour from the previous day, and the same hour from the previous week as separate feature channels. This strategy is simple and often effective, but it depends on fixed assumptions about what temporal anchors matter most. A more flexible family of approaches learns these temporal scales directly using parallel encoders, hierarchical recurrent modules, dilated convolutions, or attention over multiple resolution levels. The key insight is that temporal scale is not just an engineering hyperparameter; it is a structural property of the forecasting problem [43], [47].

For taxi demand in particular, multi-scale methods are especially appropriate because the service is highly responsive to recurring human routines. The same district can behave very differently on weekday mornings, Friday evenings, weekends, and holiday periods. Moreover, the intensity of these patterns may vary across zones. A business district may show strong daily periodicity, while an airport may show broader multi-day or weekly variation linked to travel patterns. This means a robust model should not assume that all temporal information is equally informative for all nodes at all times [20], [25].

The project's four-scale design can be understood in this literature context as a pragmatic but theoretically coherent decomposition. The hourly branch preserves immediate recency. The daily branch captures within-day repetition and typical local rhythm. The weekly branch addresses weekday–weekend structure and broader behavioural recurrence. The monthly branch serves as a long-range trend context that can stabilise the model against short-lived noise. The literature does not prescribe exactly these four windows as universally optimal, but it strongly supports the principle behind them: short-term prediction becomes more reliable when informed by explicitly separated temporal contexts [20], [43], [47].

Another benefit highlighted by the multi-scale literature is interpretability. When a model is organised by time horizon, it becomes easier to reason about what kind of information is driving the forecast. This is especially useful in research-oriented forecasting systems, where one often wants to know whether gains come from local recency, medium-range seasonality, or longer historical context. The framework also uses stochastic sampling of branch behaviour through MC Dropout to estimate predictive uncertainty [18]. This means the project not only separates temporal scales structurally but also measures their stochastic behaviour. From a literature perspective, this is an interesting bridge between multi-scale representation learning and uncertainty analysis.

At the same time, multi-scale models introduce extra design complexity. The more branches one uses, the more important fusion becomes. A poorly designed fusion layer can negate the benefits of separate temporal encoding by collapsing incompatible information too aggressively. The literature offers several fusion strategies, including concatenation followed by projection, gating, attention-based weighting, and hierarchical fusion mechanisms [43], [48], [49]. The project currently uses linear fusion before the GRU-based output stage. This is a reasonable baseline choice, but the literature suggests that learnable cross-scale attention or adaptive gating could be explored later as a more expressive alternative [48], [49].

## 2.5 Spatial Dependence Beyond Geography

A recurring lesson in transportation literature is that "space" should not be reduced to Euclidean closeness. Two zones may border each other yet function quite differently, while zones farther apart may be tightly linked through mobility behaviour. This is especially true in taxi systems, where trip patterns often follow business corridors, airport routes, nightlife clusters, and commuting flows rather than simple neighbourhood adjacency **rong2024odsurvey**, [13]. As a result, graph construction itself becomes a research question.

The most basic graph uses shared physical borders. This is easy to build and intuitively interpretable, but it can be too coarse. Distance-based graphs soften the binary decision by weighting nodes according to geographical separation. These are useful when movement likelihood decays smoothly with distance, but they still fail to capture directed or asymmetric travel relationships. Mobility-flow graphs improve on this by using actual trip movement between regions. Such graphs can encode strong functional dependence even when geography alone would not. Similarity graphs go one step further, linking zones with correlated historical behaviour or comparable temporal signatures [13], [22].

The present project chooses an origin-destination flow matrix as the basis for graph construction. In literature terms, this is a strong choice because it ties the graph to actual travel interaction. The system maps zones through a lookup table and constructs adjacency when flow weight is positive. This means the graph is operationally grounded in the data-generating process, not merely overlaid as a generic structure. For a taxi-demand problem, this is more meaningful than an arbitrary lattice or shapefile-only adjacency **wang2024od**, [22].

However, literature also makes clear that graph construction and graph usage should be aligned. If edge weights quantify flow intensity, then message passing ideally should distinguish stronger and weaker edges rather than treating all connections as equivalent [13], [50]. The current implementation converts the dense matrix to sparse connectivity but does not make `SAGEConv` explicitly use edge weights, even though confidence-derived edge weights can be constructed. This gap between graph semantics and graph computation remains one of the clearest methodological improvement points. It suggests at least three literature-grounded future directions: weighted graph convolution, dynamic edge reweighting based on time or confidence, and joint learning of relation strength rather than fixed thresholding [13], [51], [52].

The literature on dynamic graphs is also relevant here. Urban mobility relationships are not constant over time. Morning commute patterns differ from evening leisure travel; weekday relations differ from weekend relations. A static OD graph captures average structure but not temporal variation in connection strength. This does not mean the current static graph is wrong, but it does mean that the thesis should frame it as an approximation. Dynamic graph models, time-conditioned edge weights, or multi-graph ensembles could later provide richer spatial realism [51], [52], [53].

## 2.6 Graph Neural Networks for Transportation Forecasting

Graph neural networks became central in transportation forecasting because they allow node representations to be updated through learned neighbourhood aggregation. In road traffic, the intuition is straightforward: nearby sensors influence one another through network flow. In taxi demand forecasting, the same principle applies more abstractly at the zone level: regions linked through travel behaviour or proximity may share demand dynamics or error structure. GNNs convert this intuition into learnable operators.

Early graph convolutional methods generally relied on spectral formulations or fixed graph filters. Later spatial GNNs, including Graph Convolutional Networks, Graph Attention Networks, and GraphSAGE, shifted focus toward more flexible neighbourhood aggregation. Graph Attention Networks learn to weight neighbours through attention coefficients. GraphSAGE learns how to aggregate neighbourhood features and is often favoured for scalability and inductive behaviour. In transportation literature, no single GNN variant is universally best; performance depends heavily on graph type, feature design, horizon, and training protocol.

The current project's use of GraphSAGE is methodologically sensible because the graph stage receives heterogeneous node features rather than raw graph signals alone. Node inputs include the base temporal prediction, a confidence or weight term, and optionally historical mean features. The task is then to infer a residual correction. This is a good fit for GraphSAGE because the model can aggregate neighbour attributes and infer how the local node should be adjusted relative to its relational context. The graph model is lightweight, uses neighbourhood aggregation, nonlinear activation, dropout, and predicts a scalar residual that is added back to the base prediction. This places the model within a well-established class of graph refinement architectures.

The loss design is also noteworthy. Rather than minimising absolute prediction error directly on raw demand, the graph model optimises residual error with sample weighting. If confidence is provided, node losses are weighted accordingly after normalisation. This is an elegant compromise between full graph forecasting and purely heuristic smoothing. It allows the temporal model to carry most of the predictive burden while the graph stage focuses on structured correction where it is most reliable.

Still, the GNN literature suggests several comparisons that would strengthen the thesis. First, GraphSAGE should ideally be compared with at least one alternative such as GCN or GAT. Second, If edge weights encode flow intensity, message passing should ideally be sensitive to differences in edge strength rather than collapsing all connected pairs into a uniform neighbourhood relation. Third, graph depth should be controlled to avoid over-smoothing, especially since Taxi Zones may vary sharply even among neighbours. Fourth, the graph objective should be evaluated under a temporally valid protocol to demonstrate actual forecast refinement rather than same-snapshot fitting. These issues do not undermine the current choice of GraphSAGE, but they define the standards against which its use should be discussed in the literature review and later experiments.

## 2.7 Two-Stage Residual Refinement Versus End-to-End Spatiotemporal Learning

One of the most important conceptual distinctions in the literature is whether spatial and temporal modelling should be tightly intertwined or sequentially decomposed. End-to-end spatiotemporal models often dominate benchmark leaderboards because they jointly optimise all components. However, these models also make it harder to identify what each part contributes. If performance improves, is it because temporal encoding became better, because spatial propagation worked well, or because one compensates for another in a way that is hard to interpret [13], [14]?

Residual two-stage refinement offers a different philosophy. Instead of trying to learn everything simultaneously, it first obtains a competent temporal forecast and then learns structured corrections. This approach is common in calibration, post-processing, and boosting-style systems, even if it is less fashionable in some deep-learning benchmark literature. Its strength lies in modularity and interpretability. If the graph stage improves performance, one can directly measure the incremental value of spatial correction. If it fails, the base temporal model remains intact [15].

The present project clearly adopts this residual view. The graph refinement stage treats the temporal prediction as a prior and learns residual adjustment. This is theoretically attractive because taxi demand is already strongly predictable from local temporal history; the graph stage need not rediscover that structure from scratch. Instead, it can focus on neighbourhood-based consistency and cross-zone correction. In practice, this also reduces the burden on the graph model because residuals are often smoother and lower-variance than raw targets.

From a literature perspective, residual refinement also provides a natural bridge to ensemble thinking. The temporal model and graph model can be seen as experts with different strengths: one captures intra-zone temporal recurrence, the other captures inter-zone relational adjustment. Their combination is not a crude average but an ordered composition in which the graph stage edits the temporal output. This is more principled than simple ensembling and more interpretable than monolithic end-to-end fusion [14], [15].

The main methodological challenge, again, is evaluation. Residual refinement must be assessed under conditions that match true forecasting. If the graph stage is trained on the same nodes and same hour that it later "improves," then what is being measured may be correction fit rather than predictive generalisation. The literature is clear that temporal ordering matters in forecasting evaluation, and rolling-origin or prequential evaluation is generally preferred to leakage-prone estimation setups [54], [55]. Therefore, for this project to fully realise the promise of two-stage residual design, later experimentation should enforce graph-stage train/test separation across hours or rolling windows. This remains one of the most important improvement points in the current framework.

## 2.8 Historical Summary Features and Lightweight Context Injection

A subtle but useful idea in forecasting literature is that a downstream model does not always need access to full raw sequences if it can receive carefully chosen summary statistics. This matters because graph models are often simpler than temporal models and may become expensive if full temporal sequences are attached to every node. Historical summary features provide a compromise: they inject context compactly [13], [43].

The graph stage can receive mean demand over the previous 24 hours, 168 hours, and 720 hours. These features encode short-, medium-, and longer-term demand level without forcing the graph network to process complete time series. In literature terms, this is a form of context distillation. The temporal stage extracts rich sequential information; the graph stage then receives both the base prediction and compressed historical indicators that help judge whether a correction is plausible [32], [56].

Such features are especially useful when base predictions are noisy. Suppose two zones have the same predicted next-hour demand, but one has a high recent mean and the other has been mostly inactive. A graph model receiving recent average features can treat those cases differently. Historical features also typically require stabilisation before graph use, especially for skewed count distributions, and transformations such as logarithmic compression are well supported by the literature [32], [57].

The literature suggests that this design could be expanded. Instead of only simple means, one could include recent variance, trend slope, same-hour-last-week value, holiday flags, or uncertainty summaries. However, there is also virtue in parsimony. The current design keeps graph features compact and interpretable, which is appropriate for a first thesis implementation. This makes historical summary features a lightweight but theoretically grounded mechanism deserving empirical validation through ablation.

## 2.9 Confidence Fusion as an Emerging Research Direction

The notion of confidence fusion deserves special attention because it is one of the most distinctive aspects of the project. Confidence is not derived from one metric alone. Instead, it combines three components: a prior score based on historical data richness, a stability score derived from MC variance, and a neighbourhood agreement score computed from graph neighbours. These are then merged through a weighted logarithmic combination into a final confidence score [5], [18].

This design is interesting in relation to literature because it implicitly blends three epistemic perspectives. Historical sufficiency reflects how much evidence the model has seen for a zone. Stability reflects the model's internal consistency under stochastic perturbation. Neighbourhood agreement reflects external structural plausibility relative to connected zones. Very few standard forecasting pipelines combine all three explicitly, and even fewer use the result to weight graph loss. In that sense, the project is making a meaningful methodological move beyond generic uncertainty estimation [16], [58].

At the same time, literature invites a critical discussion of how such fusion should be justified. The chosen weights are currently fixed. This is reasonable as a first implementation, but one may ask whether all cities, datasets, or time horizons would require the same weighting. A learnable fusion network, Bayesian calibration layer, or validation-tuned weight search could potentially improve robustness. Similarly, neighbourhood agreement may be sensitive to graph construction quality. If the graph is incomplete or overly thresholded, neighbourhood consistency may partly reflect graph design limitations rather than true prediction reliability [59], [60].

Another open question is whether confidence should be static per prediction or interactive during message passing. Current use is primarily through loss weighting, but literature suggests richer possibilities: confidence could scale outgoing messages, reweight edges, gate attention, or decide whether a node should rely more on local versus neighbour information. These extensions would move the system closer to fully confidence-aware graph learning [58], [61].

## 2.10 Uncertainty Estimation and Robust Forecasting

Uncertainty estimation has become increasingly important in applied machine learning because prediction error alone does not reveal confidence. In forecasting, uncertainty can arise from noise in the data, sparsity of observations, model misspecification, limited training coverage, and genuine unpredictability in the underlying process [18], [59]. For urban taxi demand, these issues are especially significant because demand variability differs dramatically across zones and across hours [5], [16]. A quiet residential zone at 3 a.m. and a business district at 8 a.m. pose different uncertainty profiles, even if their average counts are similar over some period.

The uncertainty literature commonly distinguishes several forms of uncertainty. Aleatoric uncertainty reflects irreducible data noise, while epistemic uncertainty reflects uncertainty in model parameters or structure due to limited knowledge. In practice, deep forecasting systems often use approximations to obtain at least a useful proxy for uncertainty. Monte Carlo Dropout is one of the most widely used such methods. By keeping dropout active at inference time and performing repeated stochastic forward passes, one obtains a distribution of predictions whose mean and variance can be used as approximate measures of uncertainty. This approach is attractive because it is simple to implement, computationally manageable, and easily integrated into existing deep models [18].

The project directly adopts this strategy. Repeated stochastic sampling in the temporal model yields a predictive mean and standard deviation, and branch-level stochastic behaviour can also be analysed across temporal scales. Literature supports this use of MC Dropout as a pragmatic method for uncertainty-aware forecasting, especially when the objective is not full Bayesian inference but operational reliability assessment [5], [18].

What makes the current project more interesting is that uncertainty is not used in isolation. Instead, it is fused with two further signals: historical sufficiency and neighbourhood consistency. Historical sufficiency acts as a prior, rewarding zones with richer observation history. Neighbourhood consistency compares a zone's prediction with the behaviour of its graph neighbours. This confidence fusion mechanism is an important step beyond standard literature practice. Many studies compute uncertainty but do not subsequently integrate it into downstream learning. Here, uncertainty contributes directly to node confidence, and node confidence in turn weights the graph loss, reducing the influence of unreliable regions. This moves the framework from passive uncertainty reporting to active confidence-aware modelling [59].

There are, however, several literature-based questions that this design raises. First, is the confidence weighting well calibrated? MC Dropout variance is useful, but it is not guaranteed to match true predictive reliability without calibration analysis [18], [62]. Second, do the three components deserve fixed manually chosen weights, or should the weights be learned? Third, is neighbourhood consistency genuinely independent evidence, or does it partially reuse the same graph structure later employed in refinement, potentially introducing circularity? Fourth, can confidence be used not only in the node-level loss but also in message passing, edge weighting, or feature gating? These are exactly the kinds of research gaps that literature review should highlight, because they situate the project as a strong practical design but not a final solution [59], [63].

Another relevant strand of literature concerns robust learning under noisy or sparse supervision. Confidence-aware weighting resembles ideas from curriculum learning, self-paced learning, heteroscedastic regression, and sample reweighting. The common principle is that not all observations should contribute equally to optimisation [64], [65]. In the context of graph refinement, this is especially valuable because a bad node can influence not only its own error term but also the neighbourhood information used by others. Weighting the graph loss by confidence is therefore more than a local regularisation trick; it is a strategy for controlling the propagation of unreliable information in structured prediction [58], [63].

## 2.11 Rolling Forecasting, Persistent Models, and Incremental Learning

A large part of forecasting literature focuses on static train/validation/test splits. While this is useful for benchmarking, it does not always reflect how deployed systems operate. In practice, many transportation platforms need forecasts repeatedly and continuously as time advances. This motivates rolling forecasting, where the target time shifts step by step and the model is reused, updated, or retrained as new data becomes available. Rolling protocols provide a more realistic picture of operational performance because they expose temporal drift, changing uncertainty, and the practical cost of repeated prediction [54], [55].

The present project places rolling prediction at the centre of the overall framework. In the implementation, the target timestamp advances hour by hour, predictions are generated for each Taxi Zone, confidence scores are recalculated, graph refinement is applied, and both per-hour and overall metrics are saved. This design is highly consistent with deployment-aware forecasting practice and should be regarded as a genuine strength of the work. Rather than evaluating a single static fit, the thesis studies a forecasting pipeline intended to mimic repeated operational use under continuously advancing time [54].

Within such rolling settings, one important design choice concerns how much retraining should be performed. Full retraining at every step is often computationally expensive, particularly in large multi-zone systems. Persistent models, which are trained once and then reused, offer a much cheaper alternative, but they may gradually become stale under temporal drift. Incremental learning lies between these two extremes, updating existing parameters using newly arrived data while preserving previously learned structure [66], [67]. In the present project, the repository is no longer limited to persistent checkpoint reuse alone. An incremental training variant has been added so that previously trained per-zone models can be lightly updated when new rolling windows become available. However, in the current rolling script, persistent checkpoint reuse remains the default configuration, meaning that incremental adaptation exists as an implemented extension rather than the baseline operating mode.

This distinction is important for how the framework should be interpreted academically. The persistent version emphasises computational efficiency and stable checkpoint reuse, whereas the incremental version is intended to improve adaptability under temporal drift. In long rolling horizons, this is especially valuable because urban demand patterns may evolve gradually and a fully fixed temporal model may lose stability over time. The incremental extension therefore strengthens the practical realism of the system by providing a mechanism through which newly observed information can be absorbed without the cost of full retraining at every step. From a literature perspective, the project can be understood as occupying an intermediate position between fully static checkpoint reuse and expensive repeated retraining, preserving tractability while enabling a path toward continual adaptation [54], [66].

This incremental extension is particularly relevant to the present study because it addresses one of the main weaknesses of purely persistent forecasting systems, namely the potential deterioration of stability over long rolling horizons. By allowing the temporal model to incorporate newly available observations through lightweight updates, the framework is better positioned for realistic deployment scenarios in which mobility demand evolves continuously over time. At the same time, this should be described carefully in the thesis: incremental adaptation is now supported in the codebase, but it is not yet the default experimental setting in the rolling driver script [66].

Another important issue raised in the rolling forecasting literature concerns temporal integrity and information leakage. Any forecasting pipeline that reconstructs windows, scales inputs, or derives graph-related features must ensure that future information does not leak into the current prediction step. In the present framework, the temporal forecasting stage is designed to preserve historical integrity by constructing inference windows from data available up to the forecasting context end and by fitting the scaler only on history prior to the target hour. This is methodologically sound and aligns with standard forecasting practice [54], [55].

However, the graph refinement stage still raises a more cautious generalisation question. The GraphSAGE module learns residual corrections from the same hourly batch on which refined predictions are then evaluated, and although confidence-weighted losses are applied, this protocol may overestimate the true cross-temporal generalisation gain of the graph stage. In addition, the current GraphSAGE implementation uses standard `SAGEConv` layers over the graph topology, while confidence is injected through node-level weighting rather than explicit edge-weighted message passing. As a result, the temporal forecasting stage is close to a valid rolling protocol, but the graph refinement stage still requires stronger temporally separated validation if stronger claims about generalisation are to be made. This asymmetry should be discussed openly in the thesis, because it determines how strong the empirical evidence really is for the contribution of the graph component [54], [55].

## 2.12 Critical Gaps in the Existing Literature and Their Relevance to the Project

A good literature review should not only catalogue prior methods; it should identify unresolved tensions that justify the thesis. In the present case, five such tensions stand out.

The first is the tension between temporal richness and local heterogeneity. Shared spatiotemporal models can exploit cross-zone information efficiently, but they may underrepresent zone-specific dynamics. Per-zone models preserve local specificity but are harder to scale and may be weaker in sparse zones. The project responds by using per-zone temporal models combined with a spatial refinement stage, effectively splitting local learning and relational correction into separate tasks [5], [43].

The second is the tension between structural sophistication and interpretability. End-to-end spatiotemporal networks can be powerful, but they are often difficult to diagnose. A two-stage residual framework is more interpretable because one can separately inspect base predictions, uncertainty, confidence, and refinement gains. This aligns with a growing literature preference for modular systems in operational contexts [14], [15].

The third is the tension between graph availability and graph utilisation. Many studies build elaborate graphs but do not fully exploit edge semantics. The current project already identifies this issue internally: OD edge weights exist but are not explicitly used in message passing. This is therefore not merely a future coding task; it corresponds to a broader literature problem of whether graph connectivity should be binary, weighted, learned, dynamic, or confidence-modulated [13], [59].

The fourth is the tension between uncertainty estimation and uncertainty usage. Prior work often stops after reporting predictive variance. The project goes further by fusing uncertainty with historical richness and neighbourhood agreement into a confidence score used in graph optimisation. This is a meaningful step, but literature suggests room for deeper treatment: calibration, learned confidence fusion, probabilistic loss design, and confidence-aware message passing remain open [16], [59].

The fifth is the tension between benchmark evaluation and real deployment realism. The project's rolling pipeline is well aligned with practical literature, especially after incorporating incremental temporal updating for long-horizon forecasting. At the same time, the graph stage still needs a stronger temporally separated evaluation protocol if it is to claim generalisable improvement. Thus, the principal methodological gap is no longer the absence of incremental adaptation in the temporal stage, but the need for more rigorous validation of graph-based refinement under temporally realistic generalisation settings [54], [55], [66].

## 2.13 Synthesis: Positioning the Present Study

Taken together, the literature suggests that an effective urban taxi demand forecasting system should satisfy several requirements simultaneously. It should capture nonlinear temporal dependence across multiple timescales; represent spatial interaction in a graph-compatible form; remain robust to data sparsity and regional heterogeneity; provide some measure of predictive reliability; and function under a realistic rolling prediction protocol [5], [13], [43], [54]. Few simple methods satisfy all of these requirements at once. Traditional statistical models offer interpretability but weak nonlinear and spatial representation. Pure temporal deep models improve sequence learning but ignore relational correction. End-to-end spatiotemporal networks can model both dimensions jointly, but often at the cost of modularity, interpretability, and deployment flexibility [14], [15]. Uncertainty estimation is frequently included only as an output statistic rather than a driver of later learning decisions [59].

The present study occupies a distinct niche within this landscape. It proposes not an entirely new primitive model class, but a coherent combination of ideas motivated by gaps in prior work. Its temporal stage is multi-scale and explicitly tailored to zone-level demand heterogeneity. Its uncertainty mechanism is operationalised into confidence scores rather than left unused. Its graph stage performs residual correction over an OD-based relational structure rather than predicting from scratch. Its pipeline runs in a rolling setting that combines persistent checkpoint reuse with incremental temporal adaptation, thereby strengthening long-horizon forecasting stability while maintaining computational practicality.

At the same time, the literature review makes clear that the project is not methodologically complete. To strengthen its academic contribution, later chapters should pay special attention to graph evaluation protocol, explicit edge-weight usage, and ablation over confidence components, while also assessing how incremental updating contributes to long-horizon forecasting stability [54], [55], [59]. These are not peripheral enhancements; they are the most direct ways in which the project can move from a strong engineering pipeline toward a more rigorous research contribution.

In summary, the reviewed literature provides a strong rationale for the architecture chosen in this thesis. It supports the move from traditional zone-wise forecasting to deep multi-scale temporal learning, from isolated local prediction to graph-based spatial refinement, from point prediction to confidence-aware modelling, and from static evaluation to rolling deployment-style assessment with continual adaptation [13], [43], [54]. It also provides a principled basis for critiquing the current implementation and defining the next methodological questions. The thesis therefore stands on firm conceptual ground: it is responding to genuine open issues in spatiotemporal forecasting rather than assembling components arbitrarily. The following methodology chapter will build on this literature context by formalising the problem and detailing how the proposed two-stage, confidence-weighted multi-scale framework is constructed.

# Chapter 3

# Methodology

## 3.1 Problem Formulation

An example equation:

$$y = f(x) \tag{3.1}$$

## 3.2 Model Architecture

You can insert figures like this:

# Chapter 4

# Implementation

## 4.1 System Setup

Describe your implementation details here.

## 4.2 Code Example

Example Python code:

```
import torch
import torch.nn as nn

class MyModel(nn.Module):
def __init__(self):
super().__init__()
self.linear = nn.Linear(10, 1)

def forward(self, x):
return self.linear(x)
```

Listing 4.1: Example Python code

# Chapter 5

# Experiments

## 5.1 Experimental Setup

Describe dataset, metrics, baselines, etc.

## 5.2 Results

Example table:

Table 5.1: Comparison of model performance

| Model | MAE | RMSE |
|---|---|---|
| Baseline | 12.34 | 18.56 |
| Proposed Method | 10.21 | 15.43 |

# Chapter 6

# Discussion

Discuss strengths, limitations, and interpretation of results.

# Chapter 7

# Conclusion

Summarise the thesis and suggest future work.

# Appendix A

# Additional Materials

Additional experiments, derivations, or implementation details.

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